AQ6.1: Activity Questions 1 - Not Graded

This assignment will not be graded and is only for practice.
Level 1

1 point
Suppose $T: R^2 \rightarrow R^2$ T : R2 → R2 is a linear transformation, such that $T(x, y) = (x, 0)$T (x, y) = (x, 0). Which of the following options are correct?

- [ ] The matrix corresponding to TT with respect to the standard ordered basis of R^2 R2, i.e., $\{(1, 0), (0, 1)\}$ {(1, 0), (0, 1)}, for both the domain and co-domain is $\begin{bmatrix}1&0\\0&1\end{bmatrix}$[1001]
- [ ] The matrix corresponding to TT with respect to the standard ordered basis of R^2 R2, i.e., $\{(1, 0), (0, 1)\}$ {(1, 0), (0, 1)}, for both the domain and co-domain is $\begin{bmatrix}1&0\\0&0\end{bmatrix}$[1000].
- [ ] TT is neither one-one nor onto.
- [ ] TT is one-one but not onto.

No, the answer is incorrect.
Score: 0
Accepted Answers:
The matrix corresponding to TT with respect to the standard ordered basis of R^2 R2, i.e., $\{(1, 0), (0, 1)\}$ {(1, 0), (0, 1)}, for both the domain and co-domain is $\begin{bmatrix}1&0\\0&0\end{bmatrix}$[1000].

TT is neither one-one nor onto.

1 point
If the matrix corresponding to a linear transformation $T: R^2 \rightarrow R^2$ T : R2 → R2, with respect to standard ordered basis of R^2 R2, i.e., $\{(1, 0), (0, 1)\}\{(1, 0), (0, 1)\}$, for both the domain and co-domain, is $\begin{bmatrix}2&3\\1&4\end{bmatrix}$[2134]. Which of the following is the appropriate definition of TT?

- ○ $T(x, y) = (2x + y, 3x + 4y)$T (x, y) = (2x + y, 3x + 4y)
- ○ $T(x, y) = (2x + y, 3x - 4y)$T (x, y) = (2x + y, 3x − 4y)
- ○ $T(x, y) = (x + 4y, 2x + 3y)$T (x, y) = (x + 4y, 2x + 3y)
- ○ $T(x, y) = (2x + 3y, x + 4y)$T (x, y) = (2x + 3y, x + 4y)

No, the answer is incorrect.
Score: 0
Accepted Answers:
$T(x, y) = (2x + 3y, x + 4y)$T (x, y) = (2x + 3y, x + 4y)

Let $W = \{(x, y, z) \mid x = 2y + z\}$W = {(x, y, z) ∣ x = 2y + z} be a subspace of R^3 R3. Let $\beta = \{(2, 1, 0), (1, 0, 1)\}$β = {(2, 1, 0), (1, 0, 1)} be a basis of WW. Let $T: W \rightarrow R^2$ T : W → R2 be a linear transformation such that $T(2, 1, 0) = (1, 0)$T (2, 1, 0) = (1, 0) and $T(1, 0, 1) = (0, 1)$T (1, 0, 1) = (0, 1). Answer the questions 3, 4 and 5 using the given information.

1 point

Which of the following is the appropriate definition of TT?

- [ ] $T(x, y, z) = (x, y)$T(x, y, z) = (x, y)
- [ ] $T(x, y, z) = (x, z)$T(x, y, z) = (x, z)
- [ ] $T(x, y, z) = (y, z)$T(x, y, z) = (y, z)
- [ ] $T(x, y, z) = (x - y - z, x - 2y)$T(x, y, z) = (x − y − z, x − 2y)

No, the answer is incorrect.
Score: 0
Accepted Answers:
$T(x, y, z) = (y, z)$T(x, y, z) = (y, z)
$T(x, y, z) = (x - y - z, x - 2y)$T(x, y, z) = (x − y − z, x − 2y)

1 point

Choose the correct options.

- ○ TT is one to one but not onto.
- ○ TT is onto but not one to one.
- ○ TT is neither one to one nor onto.
- ○ TT is an isomorphism.

No, the answer is incorrect.
Score: 0
Accepted Answers:
TT is an isomorphism.

1 point

What will be the matrix representation of TT with respect to the basis ββ for WW and $\gamma = \{(1, 1), (1, -1)\}$γ = {(1, 1), (1, −1)} for R^2R2?

- ○ $\frac{1}{2}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}$21 [111−1]
- ○ $\frac{1}{2}\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$21 [1001]
- ○ $\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}$[111−1]
- ○ $\begin{bmatrix}-1 & 1\\ 1 & 1\end{bmatrix}$[−111]

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\frac{1}{2}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}$21 [111−1]

1 point

Choose the set of correct options.

- [ ] Let $u = (3, 1, 0)$u = (3, 1, 0), $v = (0, 1, 7)$, v = (0, 1, 7), and $w = (3, 0, -7)$w = (3, 0, −7). There is a linear

transformation $T: R^3 \rightarrow R^3$ T : R3 → R3 such that $T(u) = T(v) = (0, 0, 0)$T (u) = T (v) = (0, 0, 0) and that $T(w) = (5, 1, 0)$T (w) = (5, 1, 0).

☐

Let $u = (2, 1, 0)$u = (2, 1, 0), $v = (1, 0, 1)$, v = (1, 0, 1), and $w = (3, 5, 6)$w = (3, 5, 6). There is a linear transformation $T: R^3 \rightarrow R^2$ T : R3 → R2 such that $T(u) = (1, 0)$T (u) = (1, 0), $T(v) = (0, 1)$T (v) = (0, 1) and that $T(w) = (5, 6)$T (w) = (5, 6).

☐

Let $u = (1, 0, 0)$u = (1, 0, 0), $v = (1, 0, 1)$, v = (1, 0, 1), and $w = (0, 1, 0)$w = (0, 1, 0). There is a linear transformation $T: R^3 \rightarrow R^3$ T : R3 → R3 such that $T(u) = (0, 0, 0)$T (u) = (0, 0, 0), $T(v) = (0, 0, 0)$ T (v) = (0, 0, 0), $T(w) = (0, 0, 0)$T (w) = (0, 0, 0), and $T(1, 4, 2) = (2, 4, 1)$T (1, 4, 2) = (2, 4, 1).

☐

Let $u = (1, 0)$u = (1, 0), and $v = (0, 1)$, v = (0, 1),. There is a linear transformation $T: R^2 \rightarrow R^2$ T : R2 → R2 such that $T(u) = (\pi, 1)$T (u) = (π, 1), and $T(v) = (1, e)$T (v) = (1, e).

No, the answer is incorrect.

Score: 0

Accepted Answers:

Let $u = (2, 1, 0)$u = (2, 1, 0), $v = (1, 0, 1)$, v = (1, 0, 1), and $w = (3, 5, 6)$w = (3, 5, 6). There is a linear transformation $T: R^3 \rightarrow R^2$ T : R3 → R2 such that $T(u) = (1, 0)$T (u) = (1, 0), $T(v) = (0, 1)$T (v) = (0, 1) and that $T(w) = (5, 6)$T (w) = (5, 6).

Let $u = (1, 0)$u = (1, 0), and $v = (0, 1)$, v = (0, 1),. There is a linear transformation $T: R^2 \rightarrow R^2$ T : R2 → R2 such that $T(u) = (\pi, 1)$T (u) = (π, 1), and $T(v) = (1, e)$T (v) = (1, e).

Level 2

1 point

Choose the set of correct options.

☐

If the Identity matrix of order 2 is the matrix representation of a linear transformation $T: R^2 \rightarrow R^2$ T : R2 → R2, with respect to the standard ordered basis for both the domain and co-domain, then it is also the matrix representation of TT with respect to any other basis ββ for both the domain and co-domain.

☐

Let TT be an isomorphism between two vector spaces, and AA be the matrix representation of TT with respect to some basis. Then the nullity of AA is 0.

☐

If AA = $\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$[0001] is the matrix representation of $T: R^2 \rightarrow R^2$ T : R2 → R2 with respect to the standard ordered basis of R^2 R2, for both the domain and co-domain, then the matrix representation of $T \circ T$T ∘ T with respect to the same bases will also be AA.

☐

If AA = $\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$[0001] is the matrix representation of $T: R^2 \rightarrow R^2$ T : R2 → R2 with respect to the standard ordered basis of R^2 R2, for both the domain and co-domain, then the matrix representation of $T \circ T$T ∘ T will be the identity matrix.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If the Identity matrix of order 2 is the matrix representation of a linear transformation $T: R^2 \rightarrow R^2$ T : R2 → R2, with respect to the standard ordered basis for both the domain and co-domain, then it is also the matrix representation of TT with respect to any other basis ββ for both the domain and co-domain.

Let TT be an isomorphism between two vector spaces, and AA be the matrix representation of TT

with respect to some basis. Then the nullity of AA is 0.
If AA = $\begin{bmatrix}0&0\\0&1\end{bmatrix}$[0001] is the matrix representation of $T: R^2 \rightarrow R^2$T : R2 → R2 with respect to the standard ordered basis of R^2 R2, for both the domain and co-domain, then the matrix representation of $T \circ T$T ∘ T with respect to the same bases will also be AA.

Let $\beta = \{(1,0),(0,1)\}$β = {(1, 0), (0, 1)} and $\gamma = \{(1,1),(1,-1)\}$γ = {(1, 1), (1, −1)} be two bases of R^2 R2. If $\begin{bmatrix}a&b\\c&d\end{bmatrix}$[acbd] is the matrix representation of a linear transformation $T: R^2 \rightarrow R^2$T : R2 → R2 with respect to ββ for both the domain and the co-domain. Answer questions 8 and 9 using the information given above.

1 point

What is the matrix representation of TT with respect to γγ for the domain and ββ for the co-domain?

○ $\begin{bmatrix}a+b & c+d\\ a-b & c-d\end{bmatrix}$[a + ba − bc + dc − d]

○ $\begin{bmatrix}a-b\\ c-d\end{bmatrix}$[ac−b−d]

○ $\begin{bmatrix}a+b & a-b\\ c+d & c-d\end{bmatrix}$[a + bc + da − bc − d]

○ $\begin{bmatrix}a+b & -a-b\\ c+d & -c-d\end{bmatrix}$[a + bc + d−a − b−c − d]

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\begin{bmatrix}a+b & a-b\\ c+d & c-d\end{bmatrix}$[a + bc + da − bc − d]

1 point

What is the matrix representation of TT with respect to γγ for both the domain and the co-domain?

○ $\begin{bmatrix}a+b+c+d & a-b+c-d\\ a+b-c-d & a-b-c+d\end{bmatrix}$[a + b + c + da + b − c − da − b + c − da − b − c + d]

○ $\begin{bmatrix}\frac{a+b+c+d}{2} & a-b+c-d\\ \frac{a+b}{2}-c-d & a-b-\frac{2}{2}c+d\end{bmatrix}$[2a+b+c+d2a+b−c−d2a−b+c−d2a−b−c+d]

○ $\begin{bmatrix}\frac{a+b+c+d}{2} & a+b-c-d\\ \frac{a-b}{2}-c-d & a-b-\frac{2}{2}c+d\end{bmatrix}$[2a+b+c+d2a−b+c−d2a+b−c−d2a−b−c+d]

○ $\begin{bmatrix}a+b+c+d & a+b-c-d\\ a-b+c-d & a-b-c+d\end{bmatrix}$[a + b + c + da − b + c − da + b − c − da − b − c + d]

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\begin{bmatrix}\frac{a+b+c+d}{2} & a-b+c-d\\ \frac{a+b}{2}-c-d & a-b-\frac{2}{2}c+d\end{bmatrix}$[2a+b+c+d2a+b−c−d2a−b+c−d2a−b−c+d]

1 point

Which of the following options are correct?

☐ If $T: R^2 \rightarrow R^2$T : R2 → R2 is a linear transformation which is both one-one and onto then T^{-1}T −1

is also a linear transformation.

If $T: R^n \rightarrow R^m$ T : Rn → Rm then the matrix corresponding to TT, with respect to the standard bases for R^n Rn and R^m Rm is of order $n \times m$n × m.

If $T: R^n \rightarrow R^m$ T : Rn → Rm then the matrix corresponding to TT, with respect to the standard bases for R^n Rn and R^m Rm is of order $m \times n$m × n.

Consider the vector space $V = \{[\begin{smallmatrix} a & b \\ 0 & c \end{smallmatrix}] | a, b, c \in R, \}$V = {[a0bc] | a, b, c ∈ R, }, that is, the set of all $2 \times 2$2 × 2 upper triangular matrices with usual matrix addition and scalar multiplication. Then there exists an isomorphism from R^3 R3 to VV .

No, the answer is incorrect.

Score: 0

Accepted Answers:

If $T: R^2 \rightarrow R^2$ T : R2 → R2 is a linear transformation which is both one-one and onto then T^{-1} T −1 is also a linear transformation.

If $T: R^n \rightarrow R^m$ T : Rn → Rm then the matrix corresponding to TT, with respect to the standard bases for R^n Rn and R^m Rm is of order $m \times n$m × n.

Consider the vector space $V = \{[\begin{smallmatrix} a & b \\ 0 & c \end{smallmatrix}] | a, b, c \in R, \}$V = {[a0bc] | a, b, c ∈ R, }, that is, the set of all $2 \times 2$2 × 2 upper triangular matrices with usual matrix addition and scalar multiplication. Then there exists an isomorphism from R^3 R3 to VV .

Check Answers

Your score is: 0/10